

Lessons from the Evolution of Foreign Exchange Trading Strategies

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Abstract: The adaptive markets hypothesis posits that trading strategies evolve as traders adapt their behavior to changing circumstances. This paper studies the evolution of trading strategies for a hypothetical trader who chooses portfolios from forex technical rules in major and emerging markets, the carry trade and U.S. equities. The results show that forex trading alone dramatically outperforms the S&P 500. But there is little gain to coordinating forex and equity strategies, which explains why practitioners consider these tools separately. In addition, a backtesting procedure to choose optimal portfolios does not select carry trade strategies until well into the 1990s, which helps to explain the relatively recent surge in interest in this strategy. Forex trading returns dip significantly in the 1990s but recover by the end of the decade and have greatly outperformed an equity position since 1998. Overall, trading rule returns still exist in forex markets—with substantial stability in the types of rules—though they have migrated to emerging markets to a considerable degree.

Keywords: exchange rate, technical analysis, technical trading, efficient markets hypothesis, adaptive markets hypothesis

JEL Codes: F31, G14, G11, G15

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Introduction

The literature on technical analysis has established that simple technical trading rules on dollar exchange rates provided 15 years of positive, risk-adjusted returns during the 1970s and 80s before those returns were extinguished (Levich and Thomas (1993), LeBaron (2002), Olson (2004)).¹ More recently, more complex and less studied rules have produced more modest returns for a similar length of time (Neely, Weller and Ulrich (2009)). Conventional explanations that rely on risk adjustment and/or central bank intervention do not plausibly justify the observed excess returns from following simple technical trading rules. Neither can data mining explain the apparent profitability of technical analysis (Neely, Weller and Ulrich (2009)).

Andrew Lo's (2004) adaptive markets hypothesis (AMH) offers a plausible explanation for this technical trading puzzle, however. The AMH posits that profit opportunities will generally exist in financial markets but that learning and competition will gradually erode these opportunities as they become known. A core principle of the adaptive markets hypothesis is that traders learn over time, adapting their behavior to changing circumstances. This suggests that one might expect an evolution of strategies and desired investment currencies. Although the long-term secular decline in profitability of a broad range of static currency trading rules in major markets has been well documented, the literature has paid much less attention to the lessons that traders might have been learning as markets evolved. In particular, there has been little research on the possibility that dynamic trading strategies might outperform simple static rules and may not have suffered from the same decline in profitability. In contrast—at least superficially—to this prediction of time varying performance technicians believe that rules that work in one market will tend to work in other markets and presumably work over time.

¹ Menkhoff and Taylor (2007) and Neely and Weller (2011) review the literature on technical analysis in the foreign exchange market from different perspectives.

Ideally, one might like to examine the evolution of technical trading strategies by directly looking at the trading records of technicians. As these data are not available, one might infer how such strategies changed by considering how a hypothetical trader would have adapted to changing market conditions using simple rules-of-thumb.

Traders face a number of practical problems as they choose strategies to maximize their welfare. How to choose rules, individually or as part of a portfolio? How to combine technical rules in foreign exchange with carry trade or equity strategies? In practice, traders must make these choices by backtesting rules on existing data. Although a number of studies have found evidence that the profitability of technical trading rules has declined over time —i.e., Levich and Thomas (1993), LeBaron (2002), Okunev and White (2003), Olson (2004) and Neely, Weller and Ulrich (2009)— there has been very little research on properties of backtesting. This paper posits a hypothetical trader who chooses trading strategies and portfolios by backtesting from a group of commonly used technical and carry-trade rules in tradable currencies (Brunnermeier et al., 2009; Jordà and Taylor, 2009; Farhi et al., 2009; Burnside et al., 2011a, 2011b; Menkhoff et al., 2011a, 2011b). Practitioners widely use both sorts of trading strategies but researchers have done little comparison between them (Menkhoff, Sarno, Schmeling, and Schrimpf (2011b)).

Specifically, we investigate whether a hypothetical trader could use past performance of trading rule-currency pairs to predicts future performance and to construct a dynamic trading strategy that is superior to individual trading rules.² To mimic the decision process of a forex trader, we construct a dynamic strategy in the following way: We start with a pool of rule-currency pairs (including carry trades) and rank them at time t according to Sharpe ratio over

² Olson (2004) dynamically selects the best moving average rule for each of 18 currencies in successive 5-year periods from 1971-2000 and then tests these in successive 5-year out-of-sample periods. He finds that returns declined from the 1970s to about zero in the 1990s. Okunev and White (2003) find that the returns generated by momentum strategies appear to have been more persistent, at least up until the end of their sample in 2000. Burnside et al (2011b) and Menkhoff et al (2011b) consider momentum and carry trade strategies.

some past time window.³ We then form portfolios of the highest ranked N rules and measure the return to the portfolio over period $t + 1$. Each period individual rule-currency pairs are re-ranked and the results of the ranking are allowed to determine the composition of the portfolio.

In addition, we investigate what such a trader would learn about diversifying his portfolio. Given the well-documented fact that currency trading rule returns typically display very low correlation with stock market returns, one might expect that a diversified portfolio combining equity with a dynamic currency trading strategy would produce substantial benefit. In fact we show that no such benefit exists, and that this is consistent with the observed levels of excess return and volatility. Given the substantially stronger performance of the dynamic currency strategy the optimally diversified portfolio contains a rather small allocation to equity and this implies that even ignoring parameter uncertainty and sampling error the resulting impact on performance is very small. This lack of benefit to active diversification is consistent with the prevalence of “compartmentalization” of forex and equity trading activities by practitioners. We also find that the superiority of the dynamic trading strategy over equity has increased over the last decade. Given that the equity premium is still generally regarded as a puzzle, the currency trading rule premium appears to be even more of one.

Presaging the results, we find that backtesting works well for our trader. We find clear evidence that past performance does predict the future: Rule-currency pairs that are more highly ranked in backtesting have higher ex post Sharpe ratios. We also find that the Sharpe ratio of the dynamic trading strategy is much superior to that of the S&P 500. The backtesting methodology is fairly robust to the selection window. Both ex ante optimal and 1/ N portfolios produce very good Sharpe ratios in every subsample, clearly exceeding those of their constituent rules. The ex

³ Given that none of the returns appear to have systematic risk, the Sharpe ratios allow one to easily compare performance from strategies with differing volatility. Therefore we focus on Sharpe ratios as our metric for rule/strategy performance.

ante optimal forex combinations are somewhat more profitable than 1/N portfolios over the whole sample. The research does, however, confirm a dip in the profitability of major investment currencies in the 1990s and a switch to emerging market currencies around 1990. In contrast, the types of rules that are chosen are fairly stable over time, with the exception of the increased importance of the carry trade from the mid-1990s.

Our experiment also sheds light on the stability of types of rules and investment currencies over time and suggests explanations for the popularity of two practices in forex markets. First, the results suggest an explanation for the separation of forex and equity trading operations: Combining technical forex positions with equity positions does not improve on the Sharpe ratios for the technical forex positions. Second, the selection strategies do not select the bilateral carry trades in the top-ranked rules until the mid-1990s. The fact that carry trade strategies did not measure up well to the best-ranked technical rules might in part explain the almost complete lack of academic interest in the carry trade before 2006.

1. Methodology

We examine the performance of portfolios of technical trading rules that are rebalanced monthly by applying a performance criterion. We use a standard pool of rules, which we consider to be representative of those that the academic literature has investigated: 7 filter rules, 3 moving average rules, 3 channel rules and a carry trade rule. Dooley and Shafer (1984) and Sweeney (1986) look at filter rules; Levich and Thomas (1993) look at both filter and moving average rules; and Taylor (1994) looks at channel rules, for example.

A filter rule generates a buy signal for a foreign currency when the exchange rate (dollar price of foreign currency) has risen by more than y percent above its most recent low. It generates a sell signal when the exchange rate has fallen by more than the same percentage from

its most recent high. Thus

$$z_t = \begin{cases} 1 & \text{if } S_t \geq n_t(1+y) \\ -1 & \text{if } S_t \leq x_t(1-y) \\ z_{t-1} & \text{otherwise} \end{cases}$$

where z_t is an indicator variable that takes the value +1 for a long position and – 1 for a short position. We denote the exchange rate at date t (\$ per unit of foreign currency) by S_t , and n_t is the most recent local minimum and x_t the most recent local maximum. The 7 filter rules have filter sizes (y) of 0.005, 0.01, 0.02, 0.03, 0.04, 0.05 and 0.1.

A moving average rule generates a buy signal when a short-horizon moving average of past exchange rates crosses a long-horizon moving average from below. It generates a sell signal when the short moving average crosses the long moving average from above. We denote these rules by vma(S, L) where S and L are the number of days in the short and long moving averages. The moving average rules are vma(1, 5), vma(5, 20) and vma(1, 200). Thus vma(1, 5) compares the current exchange rate with its 5-day moving average and records a buy signal if on the previous trading day the exchange rate was below its 5-day moving average and on the current day it lies above it.

A channel rule counsels to buy (sell) if the price exceeds (is less than) the maximum (minimum) over the previous n days plus (minus) the band of inaction (x).⁴ Thus

$$z_t = \begin{cases} 1 & \text{if } S_t \geq \min(S_{t-1}, S_{t-2}, \dots, S_{t-n})(1+x) \\ -1 & \text{if } S_t \leq \max(S_{t-1}, S_{t-2}, \dots, S_{t-n})(1-x) \\ z_{t-1} & \text{otherwise} \end{cases}.$$

We set n to be 5, 10 and 20, and x to be 0.001 for all rules.

⁴ We define the channel rule following Taylor (1994). Sullivan, Timmermann and White (1999) instead call this rule a “support-and-resistance” rule. The Sullivan, Timmermann and White (1999) definition of the channel rule is similar to Taylor (1994), but the rule is conditioned on a formed channel, i.e., the minimum and maximum over the last n days must be within a certain distance of each other.

We consider an individual currency carry trade in which the rule takes a long position if the overnight interest rate for the foreign currency is greater than the dollar rate and a short position otherwise. This is the form of bilateral carry trade examined in Burnside, Eichenbaum, Kleshchelski and Rebelo (2011a).

We thus generate a pool of 14 rules applied to 14 dollar exchange rates: British pound (GBP), Swiss franc (CHF), Australian dollar (AUD), Canadian dollar (CAD), Swedish krona (SEK), Hong Kong dollar (HKD), Singapore dollar (SGD), Korean won (KRW), Japanese yen (JPY), South African rand (ZAR), Thai baht (THB), Czech koruna (CZK), Russian ruble (RUB), and Deutschmark/euro (DEM/EUR). The series for the DEM was spliced with that for the euro after January 1, 1999.

We sort all rules with at least 500 days of data (since the beginning of the respective samples) by Sharpe ratio. There is a maximum of $(14 \times 14 =)$ 196 rules on any given day, but missing data for some exchange rates often leave fewer than half that number of rules. The ranking and rebalancing procedures are performed every 20 business days. Thus the top-ranked portfolio's returns will be generated by a given trading rule applied to a particular currency for a minimum of 20 days, at which point it may (or may not) be replaced by another rule applied to the same or a different currency.

2. Data

Table 1 shows the 14 countries whose exchange rates—noon New York City buying rates—versus the U.S. dollar were used. All the exchange rates are from the daily database at Haver. The original source is the Board of Governors of the Federal Reserve System, Foreign Exchange Rates, H.10. The DEM/USD return series was spliced with the EUR/USD return series at the date of the introduction of the euro, January 1, 1999. We take a conservative view

of the periods in which emerging markets currencies can be traded. To avoid periods in which capital controls or market disruption would have prevented actual trading, we restricted simulated trading in the Thai baht and South African rand to start on July 2, 1997 and April 3, 1995, respectively.

Most of the interest rate data came from the BIS and consisted of overnight money market rates. For several countries, overnight interbank or money market interest rate series were obtained from their central banks: Australia, Europe, South Korea, Russia, the United States, and the United Kingdom. Interest rate data for Thailand was constructed by splicing a series from the BIS with a series from the Bank of Thailand. Japan's interest rate was constructed by splicing three series, one from the Bank of Japan and two from the BIS. Switzerland, Singapore and Japanese interest rate data exhibited a few negative values, typically early in the data. We set these interest rate observations to zero for return calculations.

3. The Performance Criterion

We now turn to the measure of excess return, which is the performance criterion we use in conjunction with the Sharpe ratio both for technical trading rules and the carry trade. We first distinguish between technical trading "rules" and technical trading "strategies." Examples of a technical trading rule are a one percent filter applied to the Japanese yen (JPY) or a moving average rule $vma(5, 20)$ applied to the Hong Kong dollar (HKD). A technical trading strategy uses some selection criterion to switch between individual rule-currency pairs.

The rules and therefore also the strategies we consider switch between long and short positions in the foreign currency. We assume that some amount is held in dollars and is reinvested daily at the domestic overnight interest rate. This can be thought of as the margin held against borrowing an amount equal in value, either in dollars or the foreign currency. If a trading

rule signals a long position in the foreign currency at date t , the borrowed dollars are converted to foreign currency at the closing rate for date t and earn the foreign overnight rate. We denote the domestic (foreign) overnight interest rate by i_t (i_t^*). Then the excess return, R_{t+1} , to a long position in foreign currency is given by

$$R_{t+1} = \frac{S_{t+1}}{S_t} \frac{(1+i_t^*)}{(1+i_t)}. \quad (1)$$

We denote the continuously compounded (log) excess return by $z_t r_{t+1}$ where z_t is an indicator variable taking the value +1 for a long position and -1 for a short position, and r_{t+1} is defined as:

$$r_{t+1} = \ln S_{t+1} - \ln S_t + \ln(1+i_t^*) - \ln(1+i_t). \quad (2)$$

The cumulative excess return from a single round-trip trade (go long at date t , go short at date $t+k$), with one-way proportional transaction cost c , is

$$r_{t,t+k} = \sum_{i=1}^k r_{t+i} + \ln(1-c) - \ln(1+c) \quad (3)$$

Therefore the cumulative excess return r for a trading strategy over the period from time zero to time T is given by:

$$r = \sum_{t=0}^{T-1} z_t r_{t+1} + n \ln \left(\frac{1-c}{1+c} \right). \quad (4)$$

where n is the number of round-trip trades. Note that a trading strategy may incur transaction costs even when individual trading rules do not, and conversely. This will happen if a strategy requires a switch between two rules holding different positions but the rules themselves signal no change of position. In this case the strategy incurs a transaction cost but the individual rules do not. If on the other hand a strategy dictates a switch from a rule requiring, let us say, a long position at time t to a different rule requiring a long position in the same currency at time $t+1$,

then no transaction cost is incurred, even though one (or both) individual rules may have signaled a change of position from time t to $t + 1$.

4. Results

As a benchmark for comparison, Table 2 presents the average performance of all rules by individual currency. That is, for each exchange rate, we construct an equally weighted portfolio consisting of the 14 rules over the available data. For most currencies the net annual returns are modest—in the range 0 to 4 percent— but the Sharpe ratios are respectable, averaging 0.39. About half the exchange rates produce statistically significant net returns, and only one exchange rate, the HKD, earned a negative net excess return. Average trading frequency is modest, from about 9 to 19 trades per year.

Of course, choosing an almost random group of trading rules and currencies would not be a sensible trading strategy. Some rules may consistently outperform others or there may be variability in the level of performance, with certain rules doing well for a while and then declining. In practice, traders seek to exploit such patterns by choosing rules that “backtest” well. In other words, traders choose rules on the basis of past performance. To emulate this behavior, we construct ex ante portfolios with the simple procedure described in section 1. After an initial period of 500 business days, we commence a monthly (20-day) selection procedure in which we rank all rules according to Sharpe ratio over a selection window at the current date; then measure the performance of N ranked rules over the next month in an out-of-sample test. To investigate the impact of time variation in rule profitability, we investigated three lengths of selection windows: the full available sample, the 1000 and 500 observation periods prior to the portfolio construction date. We emphasize that all portfolios are constructed with only ex ante information, thus ensuring that a trader could have implemented the strategies. We identify

portfolio strategies according to the rank, n , of the rule. Thus the strategy corresponding to $n = 1$ selects the top-ranked rule every 20 days. The strategy corresponding to $n = 2$ selects the second-ranked rule every 20 days, and so on. Thus strategies with small values of n will switch between rules that have accumulated a history of relatively high Sharpe ratios over the whole sample. The composition of these ex ante strategies will vary with the profitability of rule-exchange rate pairs over time, as markets gradually adapt and agents arbitrage away previously profitable trading opportunities.

Table 3 details the performance of the top 10 ex ante strategies. Thus portfolio 1 describes the performance of the strategy for which each period trades are determined by the signals of the top ranked rule. Portfolio 2 describes the performance of the strategy using the signals of the second ranked rule, and so on. Over the full sample period from April 1976 to March 2010 the best ex ante strategy earns a gross annual excess return of 6.84 percent. Since the strategy trades 16.23 times a year, transaction costs lower the gross return to a net return of 6.44 percent. The associated Sharpe ratio is a healthy 0.64. Figure 1, which plots the Sharpe ratios for the top 100 ranked strategies, reveals that higher ranked strategies tend to have better net excess returns and Sharpe ratios.⁵ As rank declines return also declines, and becomes more volatile across ranks. This supports the hypothesis that the ranking and selection procedures do indeed improve performance.

Figure 2 illustrates a striking pattern of trade frequency across rank. The top-ranked strategies have the lowest trade frequency, with portfolio 1 trading only 16.23 times a year. Frequency rises rapidly to reach a maximum of 34.55 for rule 41 and then declines. Note that the trading strategies almost always trade more than individual rules because of changes made at rebalancing periods when the strategy often switches rules/positions.

⁵ A graph of net asset returns by portfolio rank is almost identical, except for scale.

We next consider the performance of the strategies over time. Figure 3 shows the net annual excess return over time for the top 5 strategies and for their average. The consistent profitability until the early 1990s emerges clearly as does the overall decline in performance in 1990. The first conclusion we can draw from this is that although a strategy of switching between rules and currencies may mitigate the 1990s decline in profitability of individual rules, it does not eliminate it. The second conclusion is that profitability returns in the late 1990s. The portfolio has positive Sharpe ratios in 12 of the 14 years from 1997 through 2010 and its average during that period is a very respectable 0.7. Third, a 1/N portfolio of the top 5 ranked strategies provides clear diversification benefits. The annual standard deviations of the individual rules ranged from 9.1 to 9.7 percent but the standard deviation of the portfolio was only 6.25 percent.

4.1 Currency portfolios and diversification

A stylized fact in the literature on technical trading in currency markets is that returns to individual rules and portfolios of rules are uncorrelated with stock returns. Since the returns are also considerably less volatile one would expect that there should be significant diversification benefits to be gained from forming a suitable portfolio combining the returns from a technical trading strategy and a stock market index. One possible approach is to consider the performance of an ex ante optimally weighted portfolio for a mean-variance investor. However, DeMiguel, Garlappi and Uppal (2007) have recently argued that the naïve 1/N allocation rule is more robust and outperforms the optimally weighted portfolio in the context of stock portfolios because means and covariances of returns are imprecisely estimated. This issue has not been investigated in the context of foreign exchange rates, however. It is therefore of interest to be able to compare the performance of naïve and optimal portfolios of rules.

We form ex ante optimal portfolios as follows. At each date t we choose the ex ante best N ($N = 10$ and $N = 50$) individual rules according to their Sharpe ratios. We calculate the mean annual excess return and the covariance matrix of the returns to these rules over the previous 500 observations. Note that this is not the same as the covariance matrix of the trading strategy returns because the identities of the rules making up the strategy change over time. So for example, if $N = 2$ and the best 2 rules according to the selection criterion at time t are “GBP filter 0.005,” “CHF vma(1,5)”, then we calculate the mean and covariance matrix for those 2 rules over the previous 500 observations. Denoting the covariance matrix by V_t and the mean return by μ_t we obtain portfolio weights

$$w_t = V_t^{-1} \mu_t . \quad (5)$$

We set negative weights to zero and scale the weight vector to sum to one. If the non-negativity constraint is not binding, then these weights maximize the Sharpe ratio of a portfolio consisting of the N rules. Next we compute the return to a portfolio consisting of the N foreign exchange rules with optimal weights over period $t + 1$. We also construct a naïve portfolio consisting of equal weights attached to each of the N rules. Using these sequences of past returns (from currency portfolios with either naïve or optimal weights) and returns on the S&P500 equity index we use the same procedure to arrive at ex ante optimal weights for the 2-asset portfolio consisting of equity and the dynamically rebalanced portfolio of trading rule returns.

In all, we construct 12 different portfolios that vary according to whether they use the top 10 or 50 trading strategies, how the foreign exchange trading strategies are constructed (optimal or naïve weights), and whether they use an optimal combination with equity, a naïve (50-50) combination with equity or no equity in the final portfolio. Table 4 displays the results for these 12 portfolios. For ease of reference we will label the various portfolios as follows:

Optimal currency-optimal equity	OO
Optimal currency-naïve equity	ON
Naïve currency-optimal equity	NO
Naïve currency-naïve equity	NN
Naïve currency-no equity	NX
Optimal currency-no equity	OX

If we wish also to distinguish between 10-rule and 50- rule currency portfolios we will write for example OO-10 or ON-50.

Both the portfolios with $N = 10$ and $N = 50$ do very well in almost all subsamples. Table 4 shows that the portfolios NO and OO have Sharpe ratios in the range from 0.88 to 0.94 over the whole sample period 1975 – 2010. The OO-10 and OO-50 portfolios have similar performance both for the full sample and for all three subsamples. Over the full sample OO-10 has a Sharpe ratio of 0.88 compared to a value of 0.94 for OO-50. However, there is no evidence of significant diversification benefit from combining the currency portfolio strategies with equity. The overall performance of OX and NX portfolios is not detectably different from that of the OO and NO portfolios. In other words, the high Sharpe ratios are attributable entirely to the currency portfolio strategies. The only subsample for which there is some evidence to the contrary is 1987 – 1998, where OX-10 portfolio had a Sharpe ratio of 0.41 whereas OO-10 portfolio had a Sharpe ratio of 0.59. Naïve combinations of currency and equity tend to do noticeably worse than the currency portfolios on their own, the only exception again occurring during the long bull market that largely coincided with the 1987 – 1998 subsample.

The absence of any detectable diversification benefit from combining the currency portfolios with equity might appear surprising in light of the fact that they show slightly negative

correlations. (The top 10 forex strategies have daily correlations between -0.06 and -0.01 with the S&P 500 total return series over the 1975-2010 sample.) But the lack of diversification benefit is perfectly consistent with the measured levels of return and volatility. Over the full sample the net returns to equity and the dynamic trading strategy OX-10 are 5.67 and 4.96 percent respectively. But the Sharpe ratios are 0.35 and 0.83 because the forex returns are much less volatile.⁶ To illustrate how such numbers translate into portfolio weights, consider an example in which equity and the dynamic strategy earn the same annual return of 5 percent and that the annual standard deviation of the equity portfolio is 15 percent, and of the dynamic strategy is 5 percent. If the two return series are uncorrelated, then the optimal portfolio weight on equity is 0.1. The Sharpe ratio of the optimally diversified portfolio is only 5.4 percent higher than that of the low volatility dynamic strategy return, however. If we were to adopt a Bayesian perspective to account for parameter uncertainty the improvement from diversification would be even smaller. The intuition for the very marginal benefit from diversification can be understood from the example above. Excess returns for the two investment strategies are fairly similar, whereas Sharpe ratios are dramatically different because equity returns are much more volatile than currency returns. This means that there are only very modest benefits to diversification even when the two return series are uncorrelated.

Another result of interest is that the OX portfolios substantially outperform equity alone. The last panel of Table 4 shows that the Sharpe ratio of the S&P 500 over the full sample is 0.35, whereas OX-10 (50) has a Sharpe ratio of 0.88 (0.94). Only over the strong (mostly) bull market of 1987 – 1998 does equity outperform the OX portfolios. The most dramatic divergence of performance occurs over the last decade 1999 – 2010 where for OX-10 (50) the Sharpe ratio is

⁶ Serban (2010) notes the superiority of Sharpe ratios from a forex strategy that combines momentum and mean-reversion elements to an equity position.

0.78 (0.56) whereas for the S&P 500 it is 0.04. In contrast to results in equity markets, there is little evidence to suggest that naïve (1/N) portfolios of forex trading strategies outperform optimal portfolios in terms of Sharpe ratios. That is, the average Sharpe ratio produced by the NO, NN and NX portfolios are almost the same as the average Sharpe ratio produced by OO, ON and OX. In contrast, optimal combinations of the forex strategies with equity (OO and NO) do seem to produce higher Sharpe ratios than the 1/N portfolios (ON and NN).

Figure 4 shows the time series of rolling Sharpe ratios for several of the top 10 strategy portfolios, both with and without equity, as well as the rolling Sharpe ratio to a buy-and-hold position in the S&P 500. The top (center) panel displays 1-year rolling Sharpe ratios from the OO-10 and OX-10 (NO-10 and NX-10) strategy portfolios from 1990 to 2010. We choose to start the graphs in 1990 because previous research (e.g., Neely, Weller and Ulrich (2009)) has dated the decline of traditional currency trading rules to approximately this year. Contrary to the general perception of the literature, forex technical trading rules tend to do much better over 2001-2010 than over 1991- 2000. This is due to the greater inclusion of emerging markets currencies in the latter sample, however. When only non-emerging currencies are used, the portfolios generally earn negative excess returns in the past decade, consistent with the literature and the results of Pukthuanthong-Le, Levich and Thomas (2007) and Pukthuanthong-Le and Thomas (2008), who find that emerging market currencies appear to provide profit opportunities to technical rules. The bottom panel of Figure 4 displays the 1-year rolling Sharpe ratios to the S&P 500. The ratios are quite variable and show no obvious trend.

4.2 Currency portfolio composition

Our findings support the view that traders could have improved on the performance of individual trading rules by implementing a simple backtest procedure to switch between different

rules at different times. In other words, rules can be reliably ranked according to expected future performance, and these rankings change over time (see Figure 1). This then raises the question of how the composition of the portfolio strategy varies. Table 5 presents the frequency with which different rules appeared in the top 5 ranked portfolios. We find that the Ch(10,.001,1) applied to DEM/EUR was the overall “winner” in that it was used 19.9 percent of the time in the top-ranked portfolio. The Ch(10,.001,1) for the GBP was the next most frequently used rule in the top portfolio, with a frequency of 13.8 percent. The carry trade on the KRW was used 7.8 percent of the time.

Moving average, filter, channel rules and the carry trade all appear among the most-used rules in the top portfolio. And both developed and emerging market currencies are represented. However, the analysis for the full sample masks substantial variation across subsamples. Some of this variation is driven mechanically by the fact that data for some emerging markets is either not available or cannot be used for certain (earlier) periods because of the presence of capital controls or other restrictions on market activity. In Table 6 we have reproduced information for the top ranked portfolio but broken down into four distinct subperiods. The GBP ch(10,.001,1) rule during the first subperiod 1973 – 1981 was very dominant, being used 71.8 percent of the time. The next most frequently used rule was the DEM/EUR vma(5,20) which was used 12.9 percent of the time. This rule continued to be popular in the second subperiod, 1982-1990, where it was used 13.3 percent of the time. But again there was one rule in the second subperiod that appeared in the top-ranked strategy far more frequently than any other, the DEM/EUR Ch(10,.001,1) rule, which the top-ranked strategy employed 74.3 percent of the time. It is not until the third subperiod 1991 – 1999 that emerging market currencies acquire a significant role. Although the JPY vma(5, 20) rule was used often, 4 of the top 5 rules were from emerging

markets, including the SGD 0.005 filter and THB vma(1,5) which were used 38.5 and 11.1 percent of the time. In the most recent subperiod 2000 – 2010 all 5 most frequently used rules involve either the KRW, RUB, or THB. In addition, the bilateral carry trade for the KRW becomes dominant in the last subperiod, being used 26.3 percent of the time.

Figure 5 shows the frequency with which various different classes of rule appeared in the top-ranked portfolios. The top panel depicts the prevalence of types of trading rules in the best 10 ex ante trading rule strategies. The bottom panel shows the difference between each raw frequency and the percentage of the total rules that the group represents. That is, the bottom panel adjusts for the fact that some of the rule groups contain more rules and therefore would have a better chance of being represented in the top 10 trading strategies. So positive (negative) numbers in the lower panel indicate that a rule group is overrepresented (underrepresented) in the top 10 ex ante trading strategies. Over the whole sample, channel rules dominate, followed by MA, small filter rules (up to 0.02), large filter rules (greater than 0.02) and the carry trade. Perhaps the most striking feature to emerge is that significant appearance of the bilateral carry trades occurs only from the mid-1990s on.⁷ The top 10 portfolio strategies use bilateral carry trades only 2.6 percent of the time over the whole sample (Figure 5).⁸ But in the last subsample the KRW carry trade becomes dominant, being used by the top ranked rule 26.3 percent of the time. It is used by the other top 10 strategies fairly frequently as well (Table 6).

The rule group prevalence seems to be reasonably stable over time with a few caveats. First, the channel rules tend to decline in importance towards the end of the 1990s, only to

⁷ The comparison of the carry trade frequency to that of the other rules is not entirely “fair” in the sense that the bilateral carry trade is only one rule while there are 3 channel rules, for example. In addition, reducing the performance evaluation window from the whole sample to 500 business days increases the representation of the carry trade in the first 10 strategies to 13.5 percent over the whole sample in this comparison. It also increases the representation of the filter rule at the expense of the channel and MA rules.

⁸ Here the bilateral carry trade is defined as a single currency strategy against a reference currency, in this case the USD. For each foreign currency the rule takes a long position if the foreign interest rate is higher than the reference currency interest rate and vice versa.

recover in the recent past. Second, as remarked above, the carry trade is unimportant until the mid-1990s.

Figure 6 shows the prevalence of exchange rates in the top 10 trading strategies. To more easily summarize the prevalence of rules over time, we divide the 15 currencies into 4 currency groups. The advanced market exchange rates consist of the GBP, CHF, AUD, CAD, SEK, JPY and DEM/EUR; the CZK and RUB are the developing European exchange rates; the HKD, SGD, KRW and TBT are the developing Asian currencies and the ZAR is the Africa group. As the composition of the exchange rate groups varies during the sample, we again normalize the frequency of each group's representation by subtracting each group's contribution to the total number of exchange rates. These statistics are in the lower panel.

Exchange rates from advanced economies dominate the top 10 ex ante trading strategies in the early part of the sample because there were few or no developing currencies in our data sample before the early to mid-1990s. In the late 1990s, currencies from developing Asian economies began to dominate the top 10 ex ante strategies and they did so until the financial crisis, when the advanced group and CZK-RUB group rose in importance. Shortening the selection period window from the whole sample to 1000 or 500 observations produces a similar pattern but there is a more rapid move to emerging Asian markets just 2 or 3 years sooner. In addition, the shorter selection windows produce greater weights on the South African rand.

5. Discussion and Conclusion

The “efficient markets hypothesis” holds that no trading strategy should be able to generate unusual profits on publicly available information—such as past prices—except by bearing unusual risk. The long-term risk-adjusted profitability of technical and carry trade strategies in the forex market suggests that the adaptive markets hypothesis would better describe market

functioning. This theory posits that traders adapt their trading strategies to changing circumstances. There has been essentially no study of how a marginal trader's strategy would have evolved over time, however. This paper studies what a hypothetical forex trader with a reasonable adaptation strategy would have learned about the possibly time varying nature of optimal rules and investment currencies, as well as diversification strategies.

We draw several conclusions from our analysis. First, a portfolio trading strategy that switches between different rule-currency pairs according to past Sharpe ratio improves substantially upon the average performance of the rule-currency pairs. That is, backtesting works because rule/currency performance is persistent. Second, there are benefits to diversifying among forex trading strategies: The optimal currency portfolio strategies (OX-10 and OX-50) considerably outperform a pure equity portfolio (S&P 500) in terms of Sharpe ratios. But the portfolio strategy optimally combined with equity in most cases produces no detectable improvement in performance compared to the portfolio strategy on its own. The naïve strategies which combined 50-50 portfolios (ON and NN) were generally inferior to the currency portfolio strategies on their own (OX and NX). The lack of a diversification benefit could help explain why firms generally treat their forex and equity positions separately. There is little point to coordinating them.

Although there has been fluctuation in the performance of the currency portfolio strategies, with a noticeable dip in the 1990s, Sharpe ratios have rebounded over the most recent decade. This observation sharply contrasts with the evidence from other studies that the profitability of individual technical trading rules had disappeared by the early 1990s. It lends support to the tentative finding in Neely, Weller and Ulrich (2009) that more complex strategies—such as those selected by an extensive backtesting exercise—experience less of a decline in performance. The

rebound in optimal rule profitability since 1998 coincides with a strong shift in the optimal strategies away from major currencies to emerging markets, first in Asia in the late 1990s and then to emerging European currencies, starting in 2004.

We find a fair amount of stability in the types of rules that optimal rule portfolios use. Channel rules decline in importance after the mid-1980s and small filter rules become more important. The most interesting change to note, however, is that the single currency carry trade comes to prominence only during the last decade (1999 – 2010). This shortly predates a surge in academic and practitioner interest in carry trade rules. The relatively poor performance of the carry trade compared to the best technical strategies prior to 1999 might explain the dearth of interest in the carry trade before the last few years.

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Table 1: Data description

Country	Currency abbreviation versus the USD	# of trading obs	Trading start date	Trading end date
UK	GBP	8664	1/2/1975	3/31/2010
Switzerland	CHF	9011	4/3/1973	3/31/2010
Australia	AUD	8337	4/7/1976	3/31/2010
Canada	CAD	8668	1/2/1975	3/31/2010
Sweden	SEK	6609	1/3/1983	3/31/2010
Hong Kong	HKD	6782	1/4/1982	3/31/2010
Singapore	SGD	4935	1/2/1990	3/31/2010
Korea	KRW	2740	5/6/1999	3/31/2010
Japan	JPY	8943	4/3/1973	3/31/2010
South Africa	ZAR	3634	4/3/1995	3/31/2010
Thailand	THB	2237	7/2/1997	12/29/2006
Czech Republic	CZK	4355	1/5/1993	3/31/2010
Russia	RUB	2373	8/1/2000	3/31/2010
Euro Area	DEM/EUR	8984	4/3/1973	3/31/2010

NOTES: The table depicts the 14 exchange rates (versus the USD) used in our sample along with the starting and ending dates of the samples.

Table 2: Average trading rule statistics by foreign exchange rate

Currency	Gross AR	Net AR	Net AR t-statistic	Sharpe	Sharpe (s.e.)	Trades per year per rule	Observations
GBP	2.37	1.99	2.24	0.36	0.16	15.24	8663
CHF	2.67	2.24	2.14	0.37	0.17	17.17	9010
AUD	1.84	1.47	1.39	0.25	0.17	14.90	8336
CAD	0.82	0.51	0.97	0.17	0.17	12.39	8667
SEK	1.98	1.58	1.38	0.27	0.19	16.22	6608
HKD	0.14	-0.10	-0.24	-0.05	0.19	9.51	6781
SGD	0.90	0.61	1.03	0.25	0.23	11.27	4934
KRW	3.31	2.96	1.70	0.46	0.27	13.88	2739
JPY	3.16	2.79	3.18	0.50	0.16	14.75	8942
ZAR	3.98	3.51	1.40	0.39	0.25	18.80	3633
THB	6.11	5.79	2.75	0.86	0.26	12.80	2236
CZK	2.31	1.89	1.28	0.32	0.24	16.78	4354
RUB	3.44	3.19	3.09	0.72	0.21	10.21	2372
DEM/EUR	3.84	3.46	3.81	0.62	0.16	15.25	8983
Mean	2.63	2.28	1.87	0.39	0.20	14.23	6161

Notes: the table presents the annual gross and net (of transaction costs) annual excess return and Sharpe ratio averaged across all 14 trading rules for each currency over the full data sample. Sample periods for different currencies are different.

Table 3: Top 10 ex ante portfolio results

Portfolio #	Gross AR	Net AR	Net AR t-statistic	Sharpe	Sharpe (s.e.)	Trades per year
1	6.84	6.44	3.73	0.64	0.17	16.23
2	5.69	5.19	3.16	0.53	0.17	19.89
3	6.64	6.17	3.66	0.62	0.17	18.79
4	5.66	5.17	2.94	0.50	0.17	19.66
5	5.38	4.85	2.84	0.47	0.17	21.14
6	3.70	3.19	1.91	0.30	0.17	20.48
7	6.02	5.48	3.38	0.57	0.17	21.70
8	6.73	6.17	3.68	0.64	0.17	22.59
9	3.86	3.29	1.95	0.34	0.18	22.88
10	5.94	5.38	3.33	0.55	0.17	22.29
Mean	5.65	5.13	3.06	0.52	0.17	20.57

Notes: the table presents for the top 10 ranked ex ante portfolio rules gross annual excess return (Gross AR), annual excess return net of transaction costs (Net AR). The sample for the ex ante portfolios is April 1976 to March 2010.

Table 4: Portfolios of technical trading rules and equity: Sharpe ratios

Construction of FX Portfolio	Combination with Equity		Top 10 ex ante rules				Top 50 ex ante rules			
			1975-2010	1975-1986	1987-1998	1999-2010	1975-2010	1975-1986	1987-1998	1999-2010
Naive	Optimal	NO	0.94	1.47	0.72	0.64	0.88	1.61	0.83	0.34
			(0.17)	(0.29)	(0.29)	(0.29)	(0.17)	(0.29)	(0.30)	(0.30)
	50-50 equity	NN	0.64	0.86	0.90	0.21	0.56	0.76	0.91	0.08
			(0.18)	(0.29)	(0.34)	(0.31)	(0.18)	(0.29)	(0.36)	(0.31)
	No equity	NX	0.91	1.55	0.52	0.66	0.84	1.69	0.58	0.23
			(0.16)	(0.30)	(0.28)	(0.27)	(0.16)	(0.28)	(0.27)	(0.30)
Optimal	Optimal	OO	0.88	1.28	0.59	0.80	0.94	1.70	0.66	0.54
			(0.16)	(0.28)	(0.28)	(0.27)	(0.16)	(0.28)	(0.29)	(0.28)
	50-50 equity	ON	0.61	0.78	0.85	0.26	0.58	0.77	0.87	0.16
			(0.18)	(0.29)	(0.34)	(0.31)	(0.18)	(0.29)	(0.36)	(0.31)
	No equity	OX	0.83	1.35	0.41	0.78	0.90	1.75	0.44	0.56
			(0.16)	(0.28)	(0.29)	(0.26)	(0.16)	(0.28)	(0.28)	(0.28)
	S&P 500		0.35	0.30	0.79	0.04				
			(0.18)	(0.29)	(0.36)	(0.31)				

Notes: the table reports Sharpe ratios with standard errors in parentheses. The trading rule portfolios consist of the top 10 and top 50 ranked rules, respectively, in the left-hand and right-hand panels. The columns labeled “Optimal equity” and “50-50 equity” show the results for 2-asset portfolios consisting of the technical trading strategies in the forex market with an S&P 500 position. The “optimal equity portfolio” uses ex ante optimal mean-variance weights; the “50-50 equity” assigns equal weights to the technical trading portfolios and the S&P 500. The “no equity” portfolio denotes the portfolios consisting of just the top N technical trading strategies. Rows labeled “naïve” weight each technical trading strategy equally; Rows labeled “Optimal” use optimal ex ante weighting on the technical strategies. The bottom panel displays the net annual excess return and Sharpe ratio to a buy-and-hold position in the S&P 500 over various samples.

Table 5: Rule prevalence over the whole sample

1			2			3			4			5		
FX rate	rule	% used	FX rate	rule	% used	FX rate	rule	% used	FX rate	rule	% used	FX rate	rule	% used
DEM/EUR	Ch(10,.001,1)	19.9	DEM/EUR	vma(5,20)	12.3	DEM/EUR	vma(5,20)	15.4	DEM/EUR	vma(5,20)	15.8	JPY	Ch(5,.001,1)	11.8
GBP	Ch(10,.001,1)	13.8	JPY	vma(5,20)	11.6	THB	vma(5,20)	7.1	JPY	vma(5,20)	12.7	DEM/EUR	vma(5,20)	10.5
SGD	filter .005	10.0	DEM/EUR	Ch(10,.001,1)	10.5	GBP	vma(5,20)	6.5	JPY	Ch(5,.001,1)	7.1	DEM/EUR	Ch(10,.001,1)	9.8
KRW	Carry Trade	7.8	THB	vma(5,20)	5.4	DEM/EUR	Ch(10,.001,1)	6.0	DEM/EUR	Ch(10,.001,1)	6.9	JPY	vma(5,20)	7.6
DEM/EUR	vma(5,20)	7.4	RUB	vma(5,20)	4.9	JPY	vma(5,20)	6.0	THB	Ch(20,.001,1)	6.0	GBP	Ch(5,.001,1)	4.7
RUB	vma(5,20)	4.7	SGD	Ch(10,.001,1)	4.2	JPY	Ch(5,.001,1)	5.6	GBP	Ch(5,.001,1)	5.1	THB	Ch(20,.001,1)	4.5
RUB	vma(1,200)	4.2	GBP	Ch(10,.001,1)	4.0	DEM/EUR	Ch(5,.001,1)	4.9	KRW	filter .03	3.8	JPY	filter .01	4.0
JPY	vma(5,20)	3.6	KRW	Carry Trade	3.6	KRW	filter .03	4.7	THB	vma(5,20)	3.6	THB	vma(5,20)	3.8
THB	vma(1,5)	3.1	GBP	vma(5,20)	3.6	JPY	Ch(10,.001,1)	4.2	KRW	Carry Trade	3.1	THB	filter .01	3.6
ZAR	Ch(5,.001,1)	2.5	GBP	vma(1,5)	3.1	THB	Ch(20,.001,1)	4.0	SEK	Ch(10,.001,1)	2.2	DEM/EUR	Ch(20,.001,1)	3.3

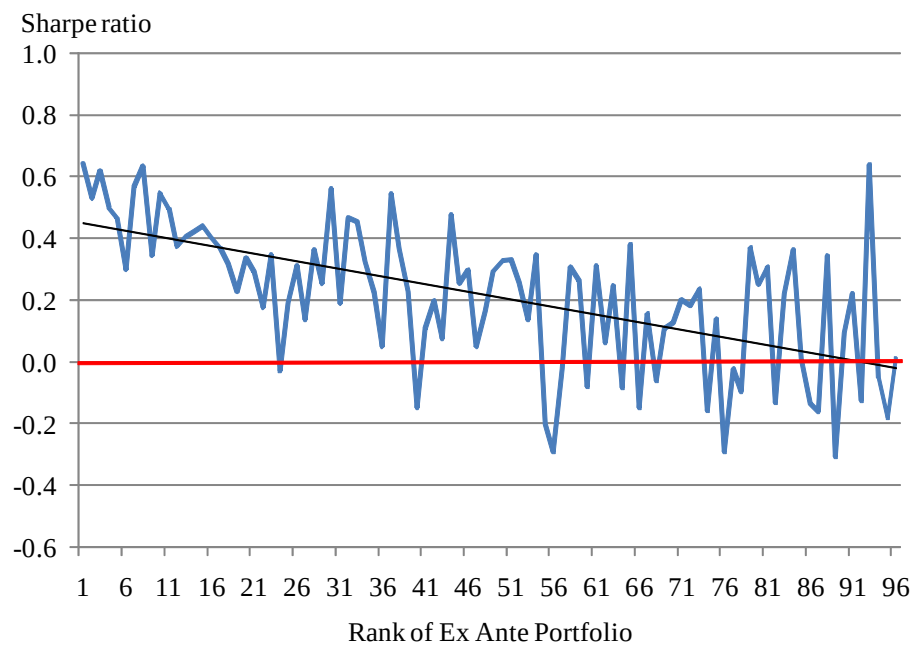
Notes: the table reports the largest 10 trading rule frequencies for the top 5 ranked ex ante portfolios over the full ample, 1973-2010. Thus the left-most columns indicate that for the strategy using the top ranked rule, Ch(10,.001,1) applied to DEM/EUR appeared 19.9 percent of the time, the Ch(10,.001,1) applied to the GBP appeared 13.8 percent of the time, and so on.

Table 6: Rule prevalence table over subsamples

1973-1981			1982-1990			1991-1999			2000-2010		
FX rate	rule	% used	FX rate	rule	% used	FX rate	rule	% used	FX rate	rule	% used
GBP	Ch(10,.001,1)	71.8	DEMEUR	Ch(10,.001,1)	74.3	SGD	filter .005	38.5	KRW	Carry Trade	26.3
DEMEUR	vma(5,20)	12.9	DEMEUR	vma(5,20)	13.3	THB	vma(1,5)	11.1	RUB	vma(5,20)	15.8
GBP	vma(5,20)	7.1	JPY	vma(5,20)	3.5	JPY	vma(5,20)	10.3	RUB	vma(1,200)	14.3
DEMEUR	Ch(10,.001,1)	3.5	GBP	Ch(5,.001,1)	2.7	ZAR	Ch(5,.001,1)	9.4	KRW	vma(5,20)	7.5
CAD	vma(5,20)	2.4	HKD	vma(5,20)	1.8	SGD	Ch(10,.001,1)	7.7	THB	Ch(20,.001,1)	5.3

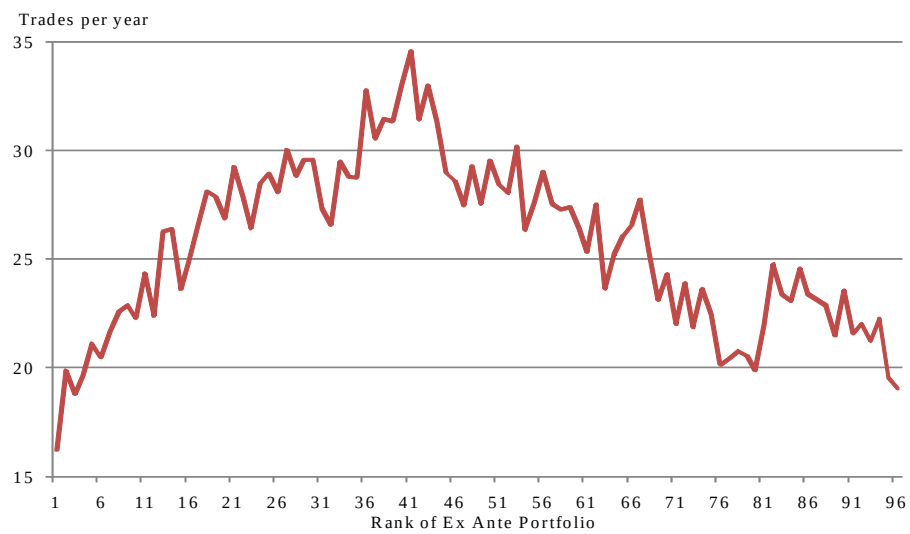
Notes: the table reports the largest 5 trading rule frequencies for the top ranked portfolio over different sample subperiods. Thus the top row entries indicate that for the strategy using the top ranked ex ante rule in the 1973-1981 subsample, the channel rule applied to the GBP with a 10 day window and a .001 band appeared 71.8 percent of the time and so on.

Figure 1



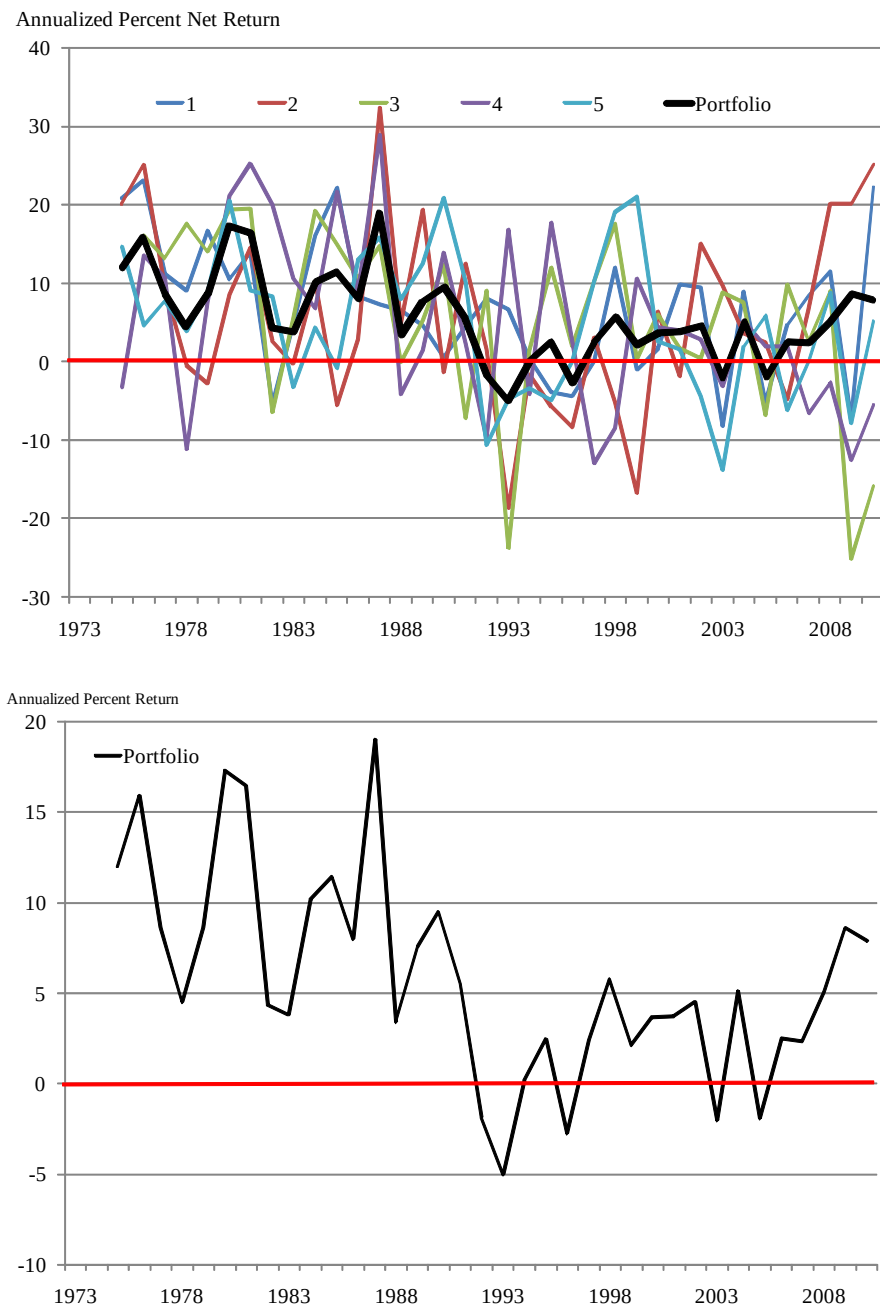
Notes: The figure displays the Sharpe ratios for the top 100 ex ante portfolio rules, along with a trendline.

Figure 2
Trades per year



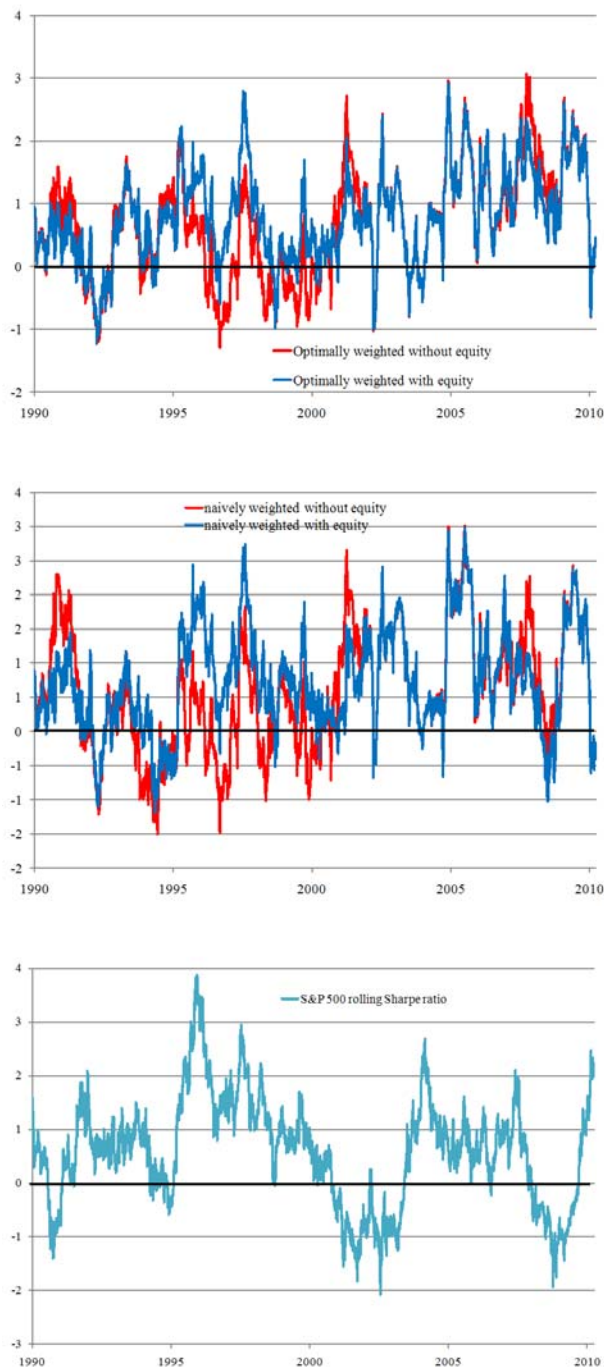
Notes: The panel displays the average number of annual trades for the top 100 ex ante portfolio rules.

Figure 3



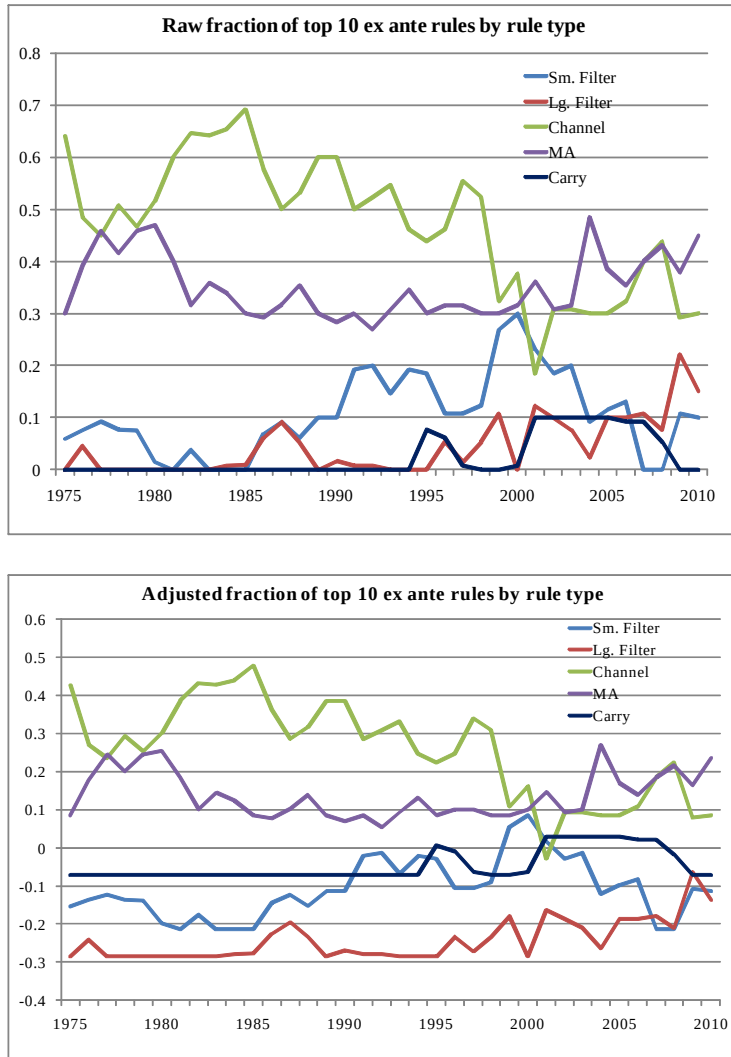
Notes: The top panel displays the net annual returns for the top 5 ex ante portfolio rules, along with the average net annual return. The bottom panel displays the average net annual return from the top 5 rules for clarity.

Figure 4: 1-year Rolling Sharpe ratios from 1990 for the top 10 strategies and the S&P 500



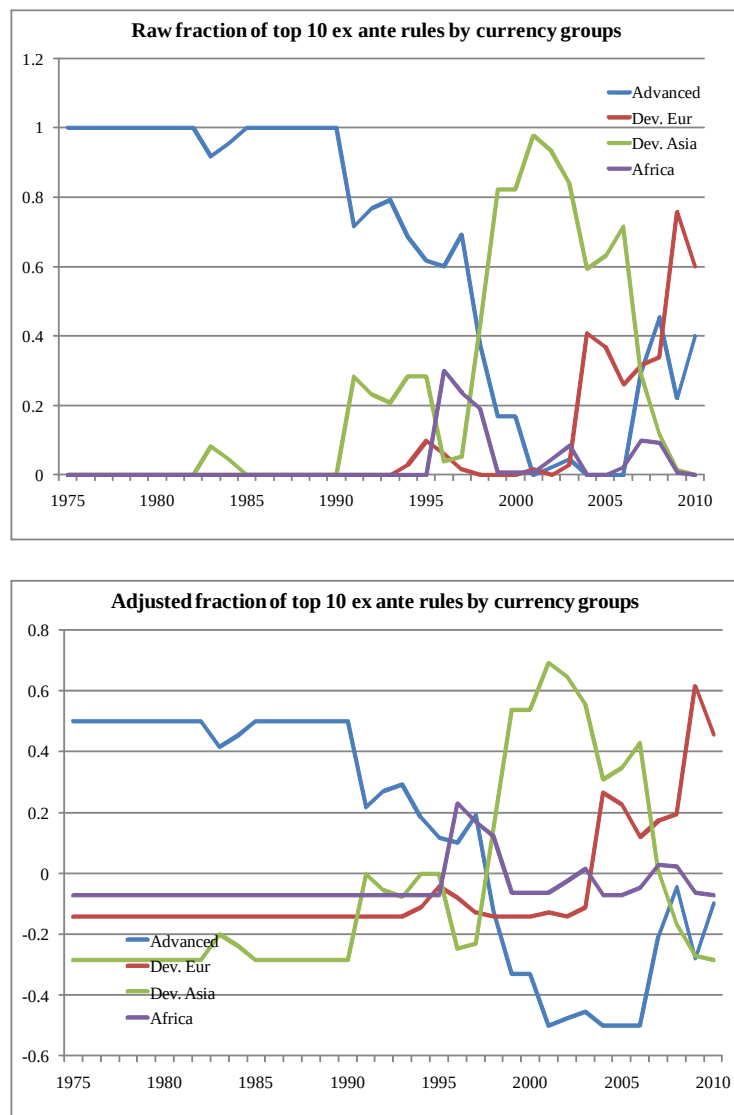
Notes: The top (center) panel displays 1-year rolling Sharpe ratios from the optimally (naively) weighted technical trading rule 10 strategy portfolios with and without optimal combination with equity, from 1990 to 2010. The bottom panel displays the 1-year rolling Sharpe ratios to the S&P 500.

Figure 5: Trading rule prevalence over time



Notes: The panels denote the prevalence of types of trading rules in the best 10 ex ante trading rule strategies. The panel on the top denotes the raw frequency of the rule groups while those on the bottom subtract from each raw frequency the percentage of the total rules that the group represents. Small filters are those less than or equal to 0.02; large filters are those greater than 0.02.

Figure 6: Exchange rate prevalence over time in the top 10 trading strategies



Notes: The panels denote the prevalence of currency groups in the best 10 ex ante trading rule strategies. The top panel illustrates the raw prevalence of each group while those on the bottom subtract from each raw frequency the percentage of the total rules that the group represents. That is, the bottom panel adjusts for the numbers of currencies in the group. The advanced market exchange rates consist of the GBP, CHF, AUD, CAD, SEK, JPY and DEM/EUR; the CZK and RUB are the developing European exchange rates; the HKD, SGD, KRW, and THB are the developing Asian currencies and the ZAR is Africa group.